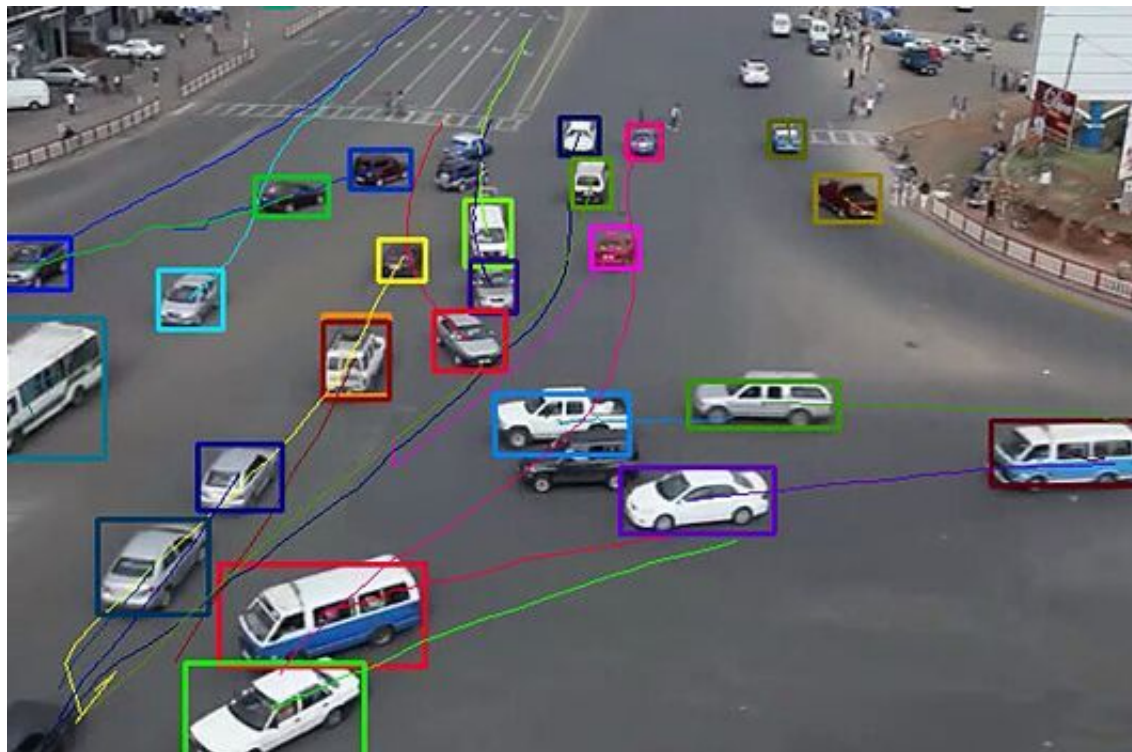


Tracking, Id, segmentation

Mark Blumenau

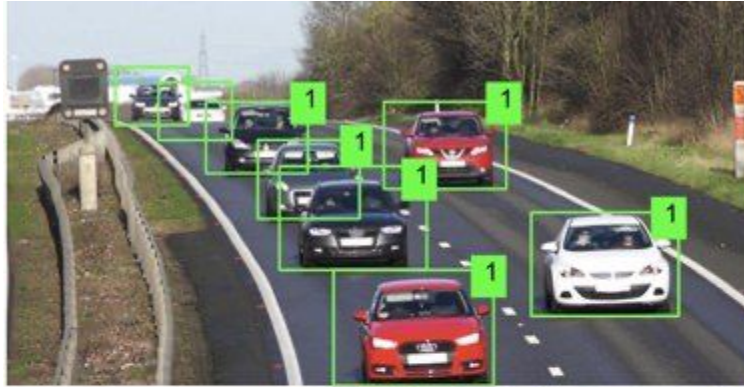
What is tracking?



What are we working with?

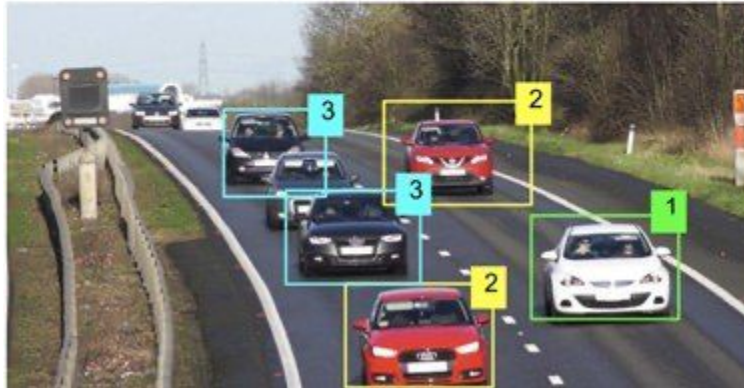


What we want?



Typical Object
Detection Algorithm

Vs.

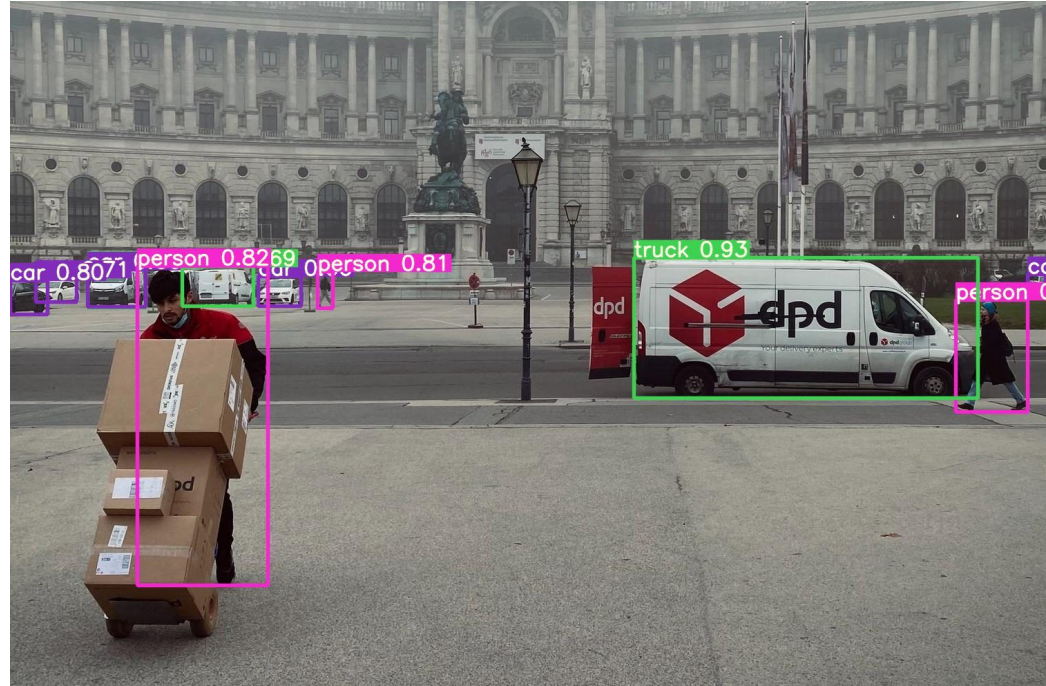


Typical Object
Tracking Algorithm

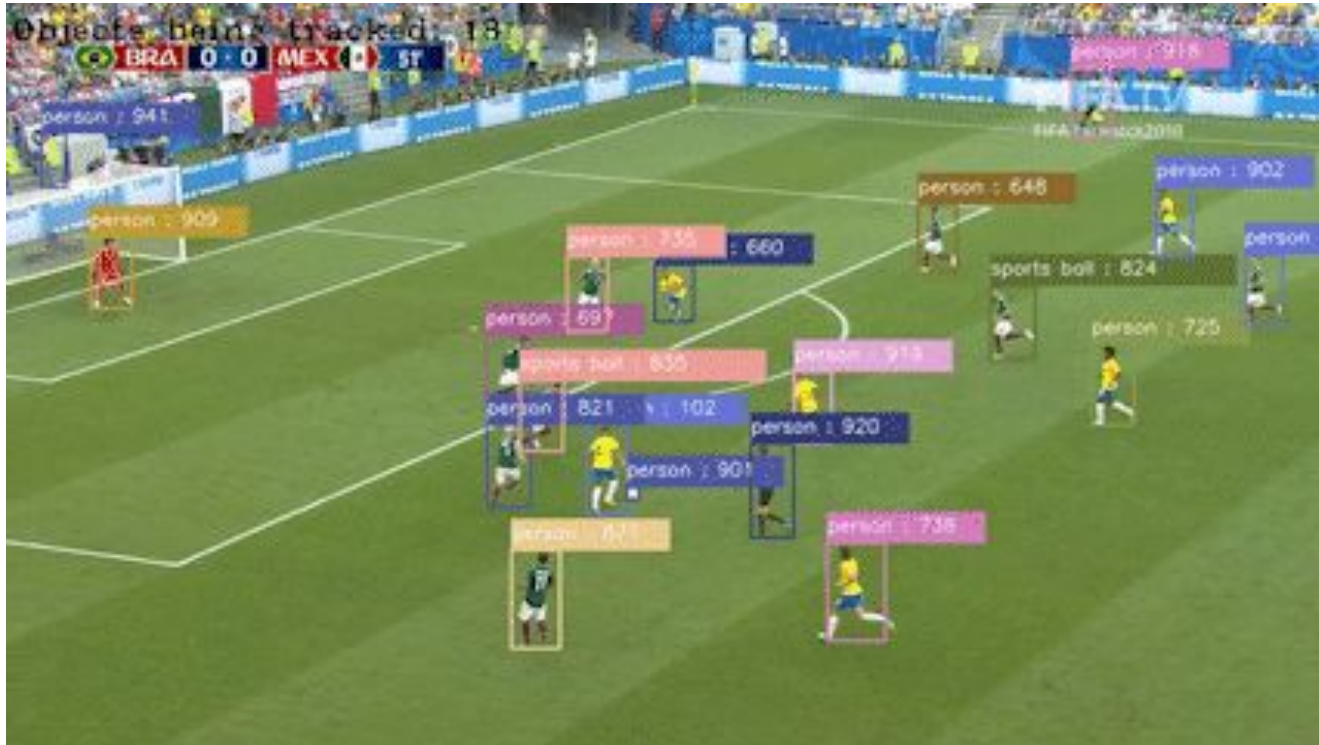
Results (output)

Why it is considered hard?

- 1) Speed :)
- 2) Background can be unstable
- 3) Perspective
- 4) Object can hide behind other objects (occlusion)

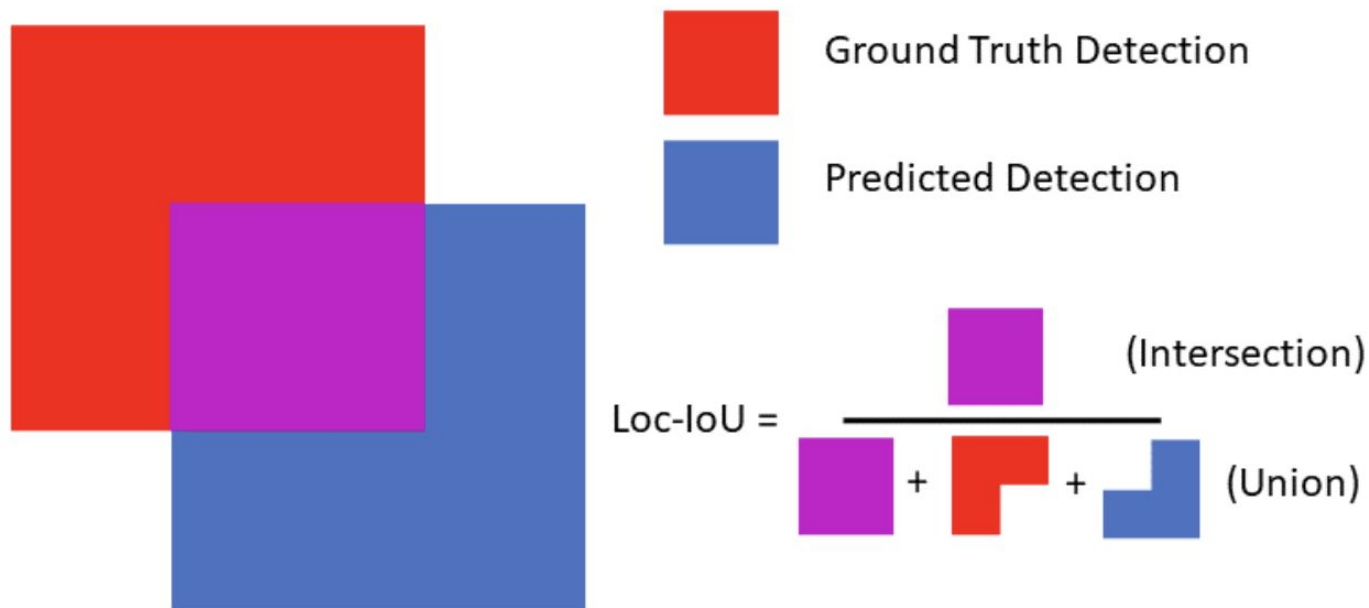


Sport! Or why it is a nightmare to work with any sport.



[GitHub - deshwalmahesh/yolov7-deepsort-tracking](https://github.com/deshwalmahesh/yolov7-deepsort-tracking): Modular and ready to deploy code to detect and track videos using YOLO-v7 and DeepSORT

The fanciest metric we have: HOTA



HOTA

$$\text{LocA} = \frac{1}{|\text{TP}|} \sum_{c \in \text{TP}} \text{Loc-IoU}(c)$$

$$\text{DetA} = \text{Det-IoU} = \frac{|\text{TP}|}{|\text{TP}| + |\text{FN}| + |\text{FP}|}$$

$$\text{Ass-IoU} = \frac{|\text{TPA}|}{|\text{TPA}| + |\text{FNA}| + |\text{FPA}|}$$

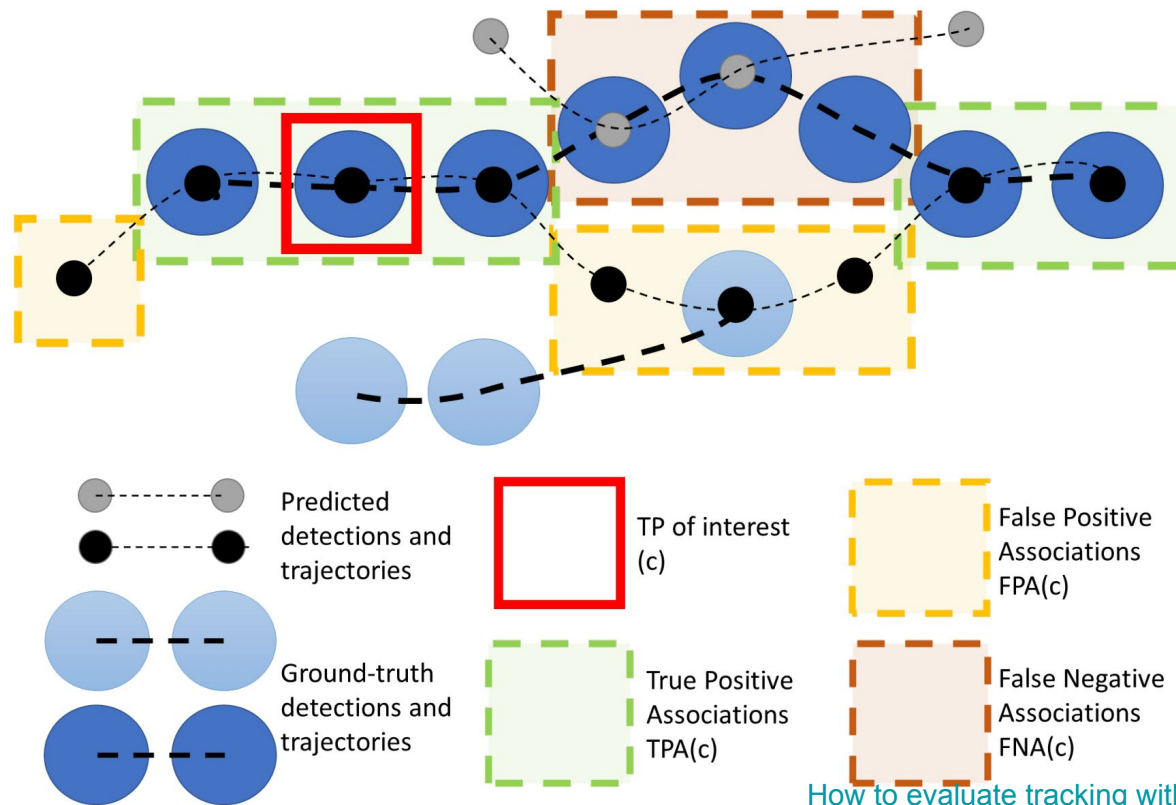
$$\text{AssA} = \frac{1}{|\text{TP}|} \sum_{c \in \text{TP}} \text{Ass-IoU}(c)$$

$$= \frac{1}{|\text{TP}|} \sum_{c \in \text{TP}} \frac{|\text{TPA}(c)|}{|\text{TPA}(c)| + |\text{FNA}(c)| + |\text{FPA}(c)|}$$

HOTA

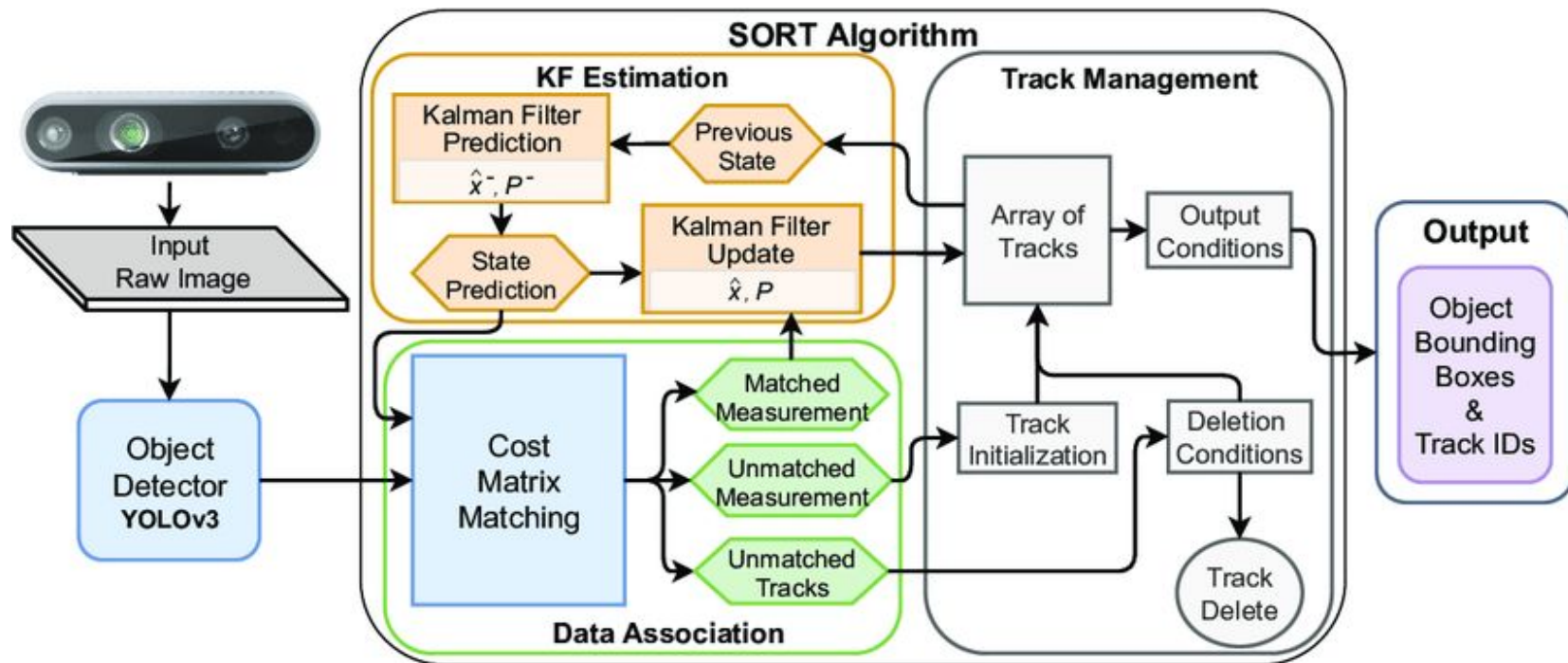
$$\begin{aligned}\text{HOTA}_\alpha &= \sqrt{\text{DetA}_\alpha \cdot \text{AssA}_\alpha} \\ &= \sqrt{\frac{\sum_{c \in \text{TP}_\alpha} \text{Ass-IoU}_\alpha(c)}{|\text{TP}_\alpha| + |\text{FN}_\alpha| + |\text{FP}_\alpha|}} \\ \text{HOTA} &= \int_{0 < \alpha \leq 1} \text{HOTA}_\alpha \\ &\approx \frac{1}{19} \sum_{\substack{\alpha=0.05 \\ \alpha+=0.05}}^{0.95} \text{HOTA}_\alpha\end{aligned}$$

A visual representation



[How to evaluate tracking with the HOTA metrics - Autonomous Vision Blog](https://arxiv.org/abs/2009.07736)

Tracking components



https://www.researchgate.net/figure/Overview-of-the-object-tracking-SORT-algorithm_fig2_358134782

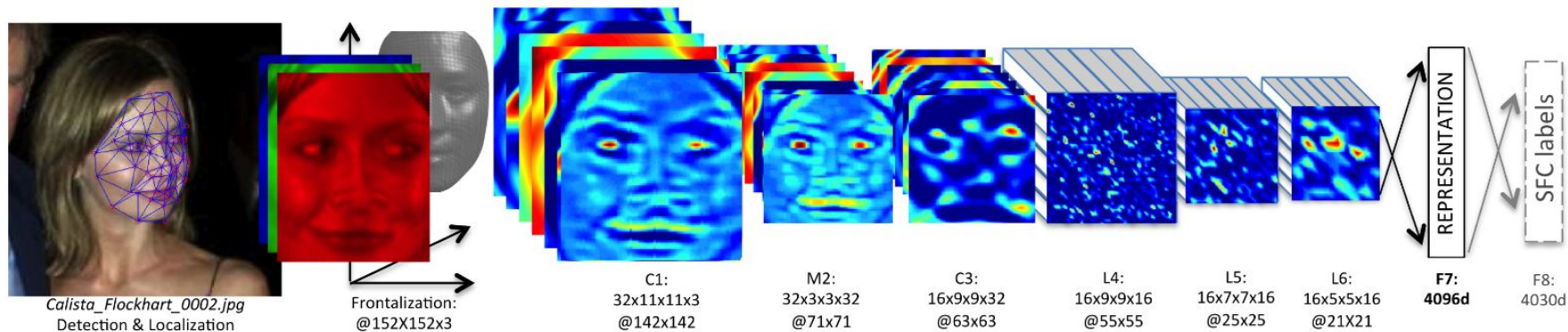
Identification (not classification!)

Labeled Faces in the Wild

~ 13 000 images and ~ 6000 people



Classic CV rules! (DeepFace)



DeepFace

We train some architecture for classification (number of classes = number of people in the data)

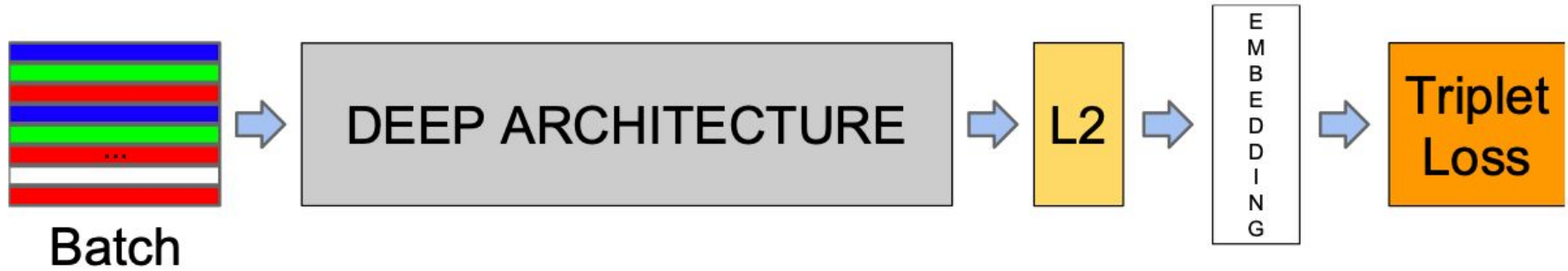
We use the outputs of the penultimate layer as a feature description of the image

The features are normalized (so that the norm is 1)

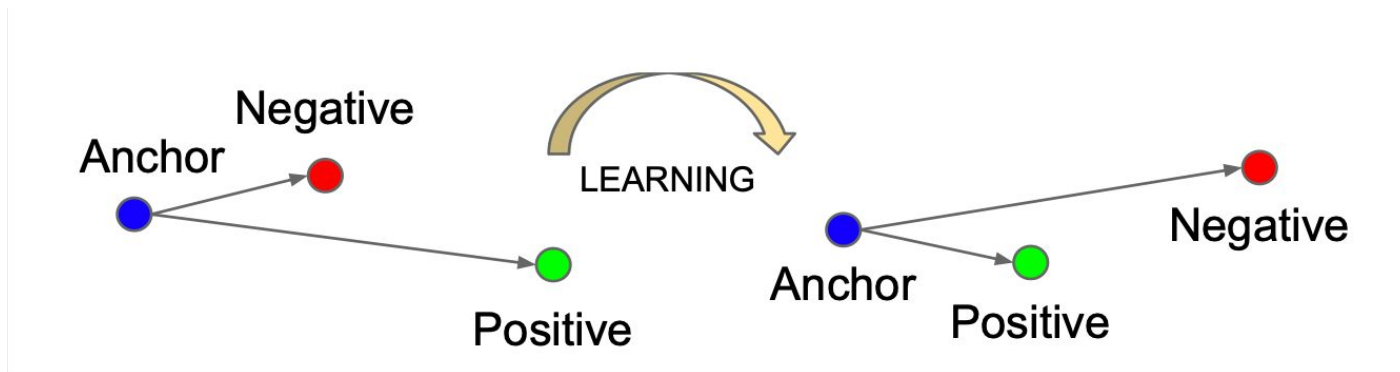
We calculate the proximity of vectors using something

FaceNet

What if we want our vectors to be close by design?



TripletLoss



$$\sum_i^N \left[\|f(x_i^a) - f(x_i^p)\|_2^2 - \|f(x_i^a) - f(x_i^n)\|_2^2 + \alpha \right]_+$$

Segmentation

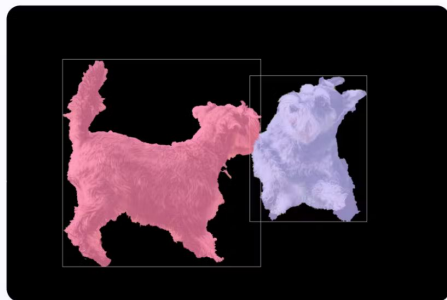
What is it?



(a) Image



(b) Semantic Segmentation



(c) Instance Segmentation



(d) Panoptic Segmentation

Metrics

Pixel based Accuracy:

$$L(y, a) = \frac{1}{n} \sum_{i=1}^n [y_i = a_i]$$

Jaccard index:

$$J_k(y, a) = \frac{\sum_{i=1}^n [y_i = k][a_i = k]}{\sum_{i=1}^n \max([y_i = k], [a_i = k])}$$

Loss function

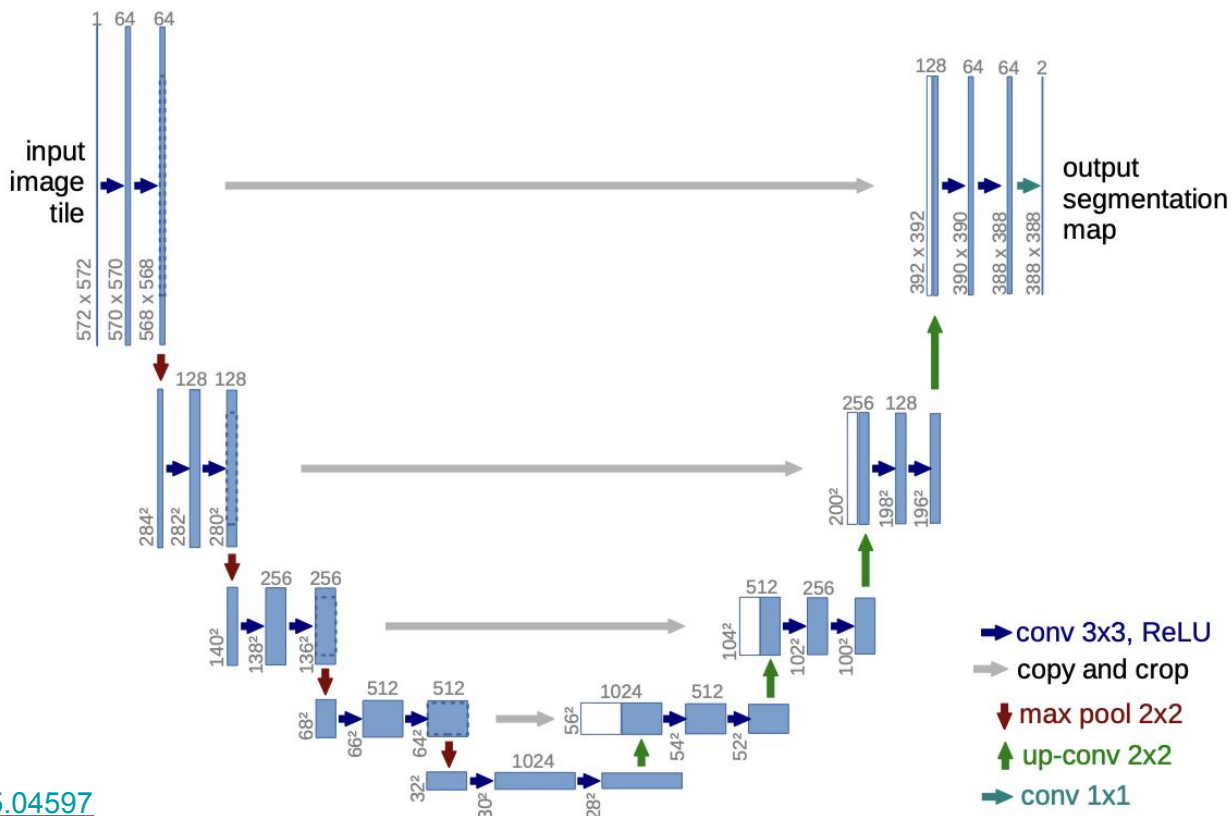
Categorical cross-entropy

$$L(y, a) = \sum_{i=1}^n \sum_{k=1}^K [y_i = k] \log a_{ik}$$

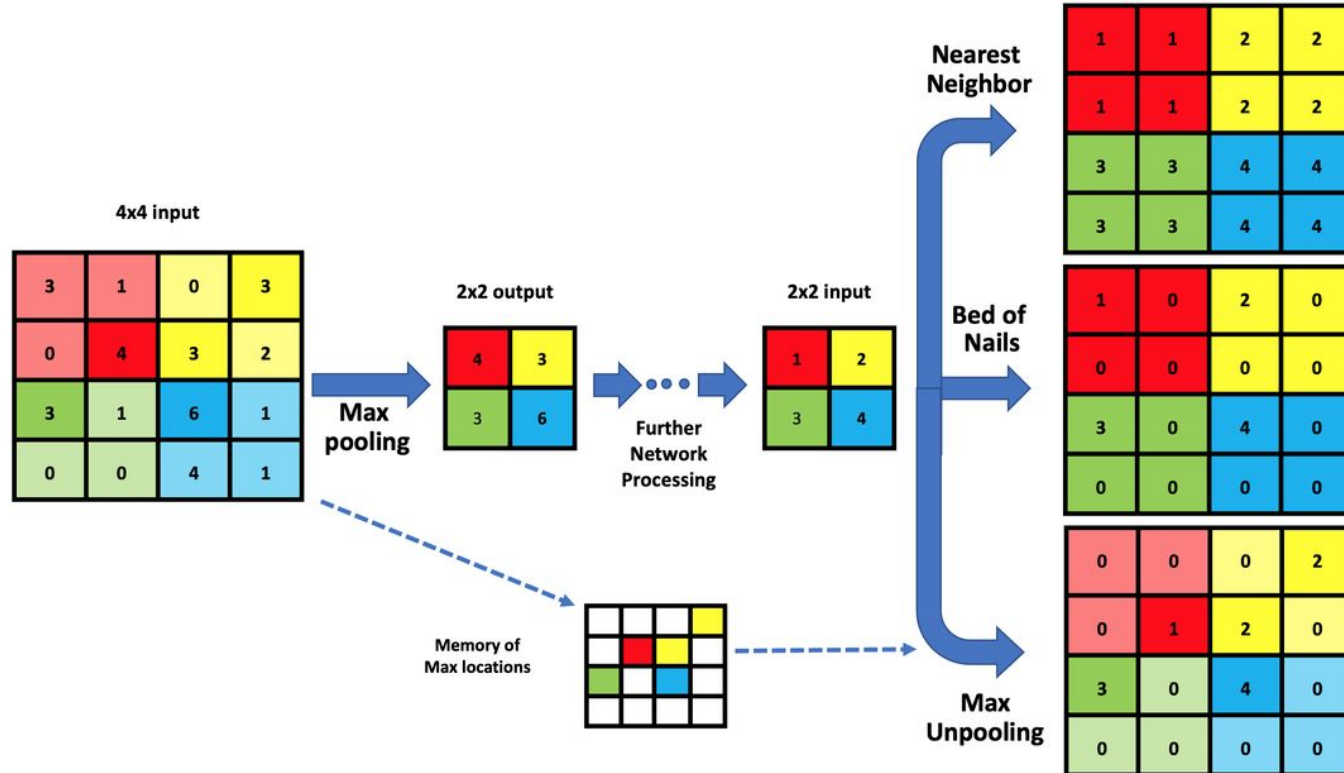
where (softmax):

$$a_{ik} = \frac{\exp(b_{ik})}{\sum_{m=1}^K \exp(b_{im})}$$

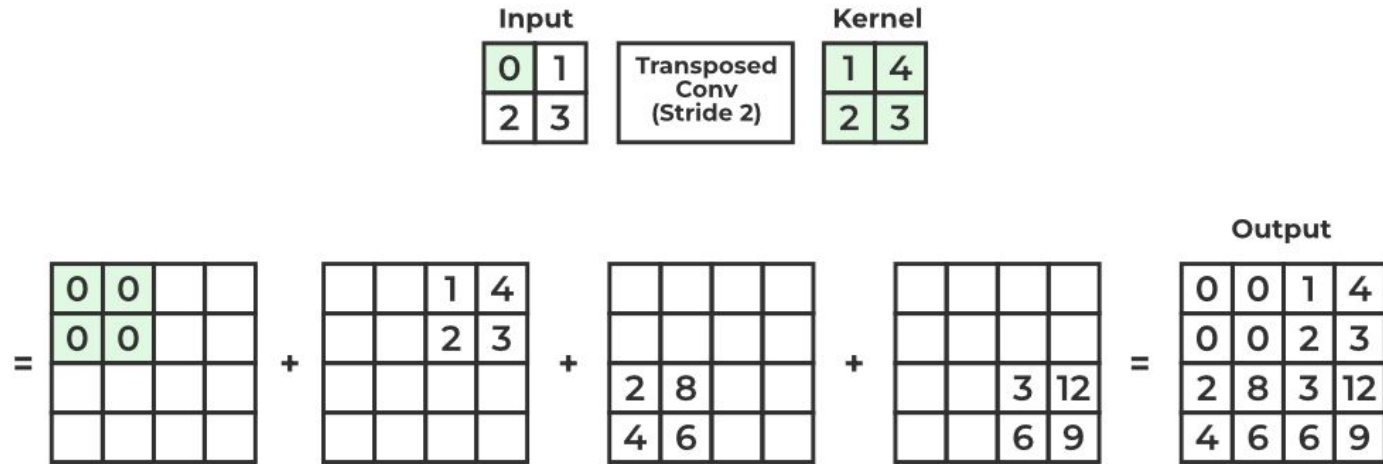
The most common arch (UNet)



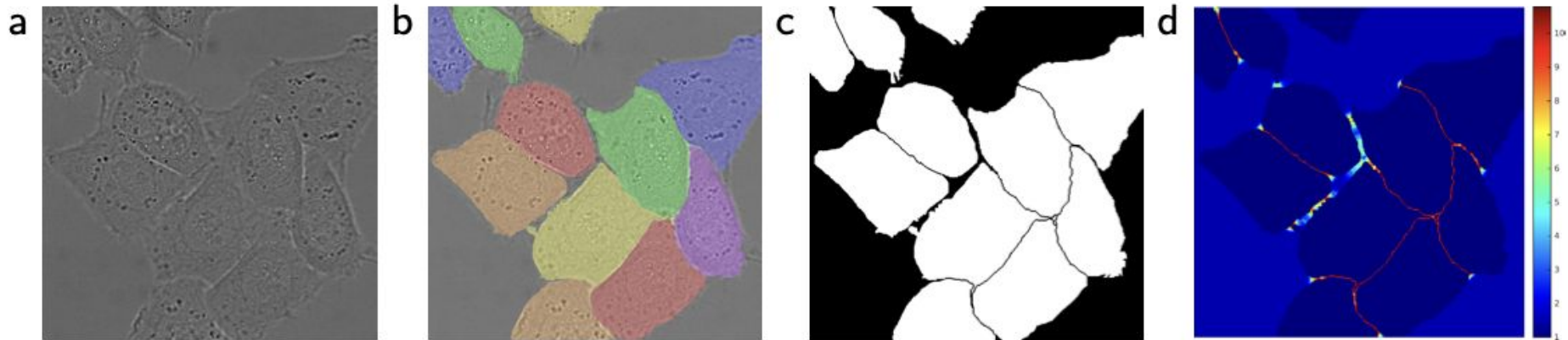
What is an upconv?



Clever upconv



Results



Mask RCNN

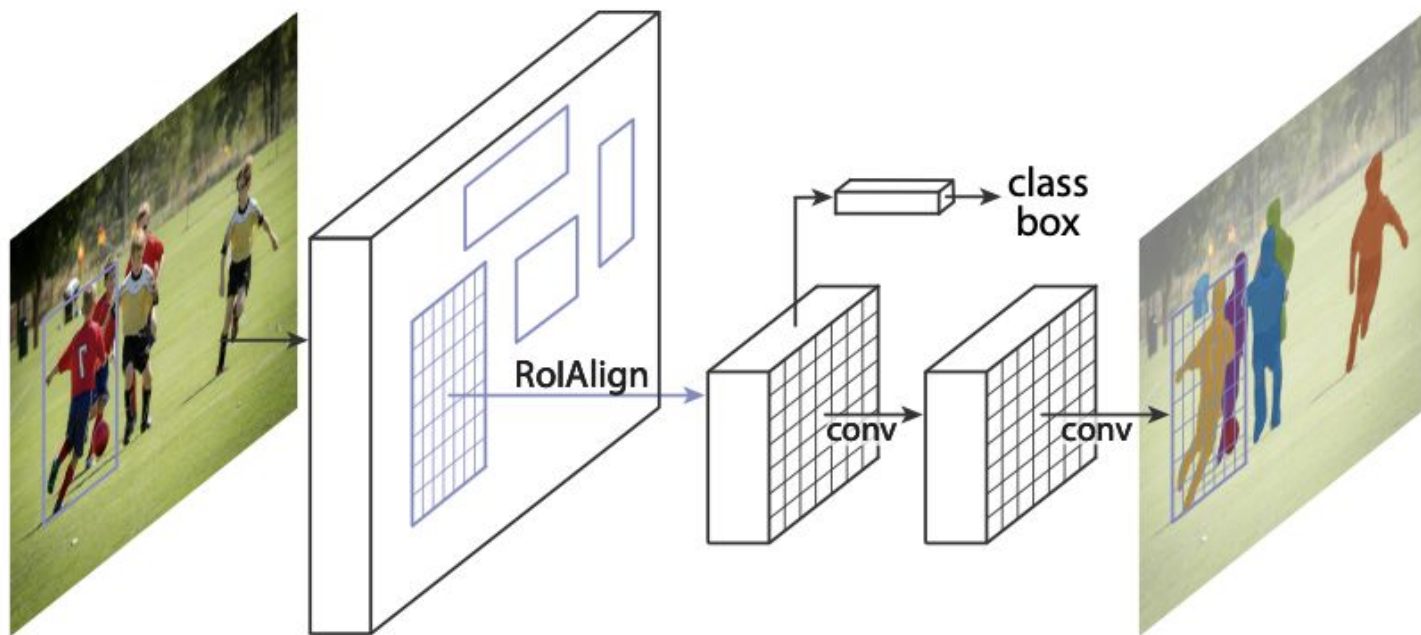
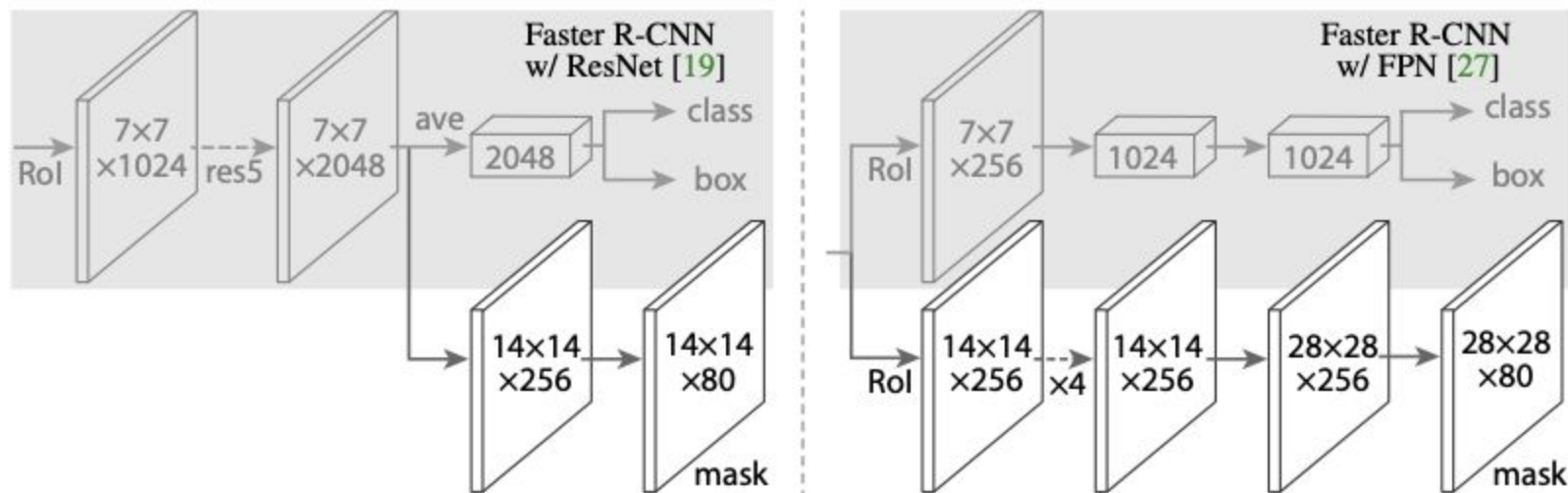


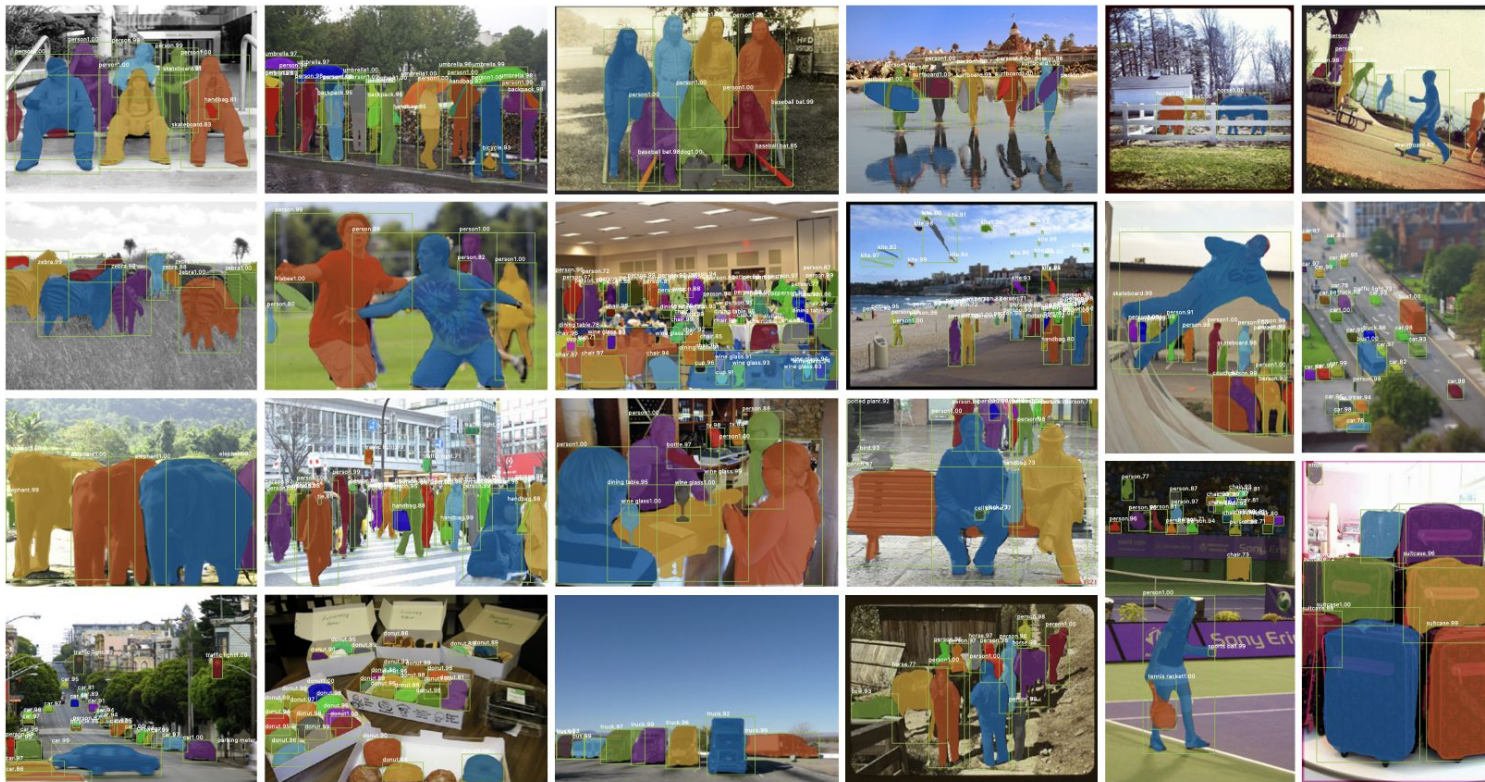
Figure 1. The **Mask R-CNN** framework for instance segmentation.

<https://arxiv.org/pdf/1703.06870>

A bit more details



Results



<https://arxiv.org/pdf/1703.06870>